FATIMA JINNAH WOMEN UNIVERSITY

Group Members:

Almas Shafqat 2023-BCS-006

Romaisa 2023-BCS-073

Sameen Noor 2023-BCS-077

Course:

Computer Architecture

Section:

BCS-5(B)

PROJECT REPORT

Project Name:

Traffic Light Controller Using Pedestrian Mode

1. Introduction

Traffic control systems are an important part of modern road transportation. They help regulate vehicle movement, reduce congestion, and improve safety for both drivers and pedestrians. In this project, a Traffic Light Controller with Pedestrian Mode is designed using Verilog and based on Finite State Machine (FSM) concepts.

The controller manages traffic signals for two roads (Road A and Road B) and also provides a pedestrian crossing signal. When a pedestrian presses a request button, the system safely stops vehicle traffic and allows pedestrians to cross.

This project practically demonstrates important Computer Architecture and Digital Logic concepts such as:

- Finite State Machines (FSM)
- Sequential vs combinational logic
- Clock-driven control units
- Registers, counters, and control signals

2. Objectives of the Project

The main objectives of this project are:

- To design a traffic light controller for two roads.
- To include a pedestrian crossing mode.
- To use FSM concepts for systematic state transitions.
- To latch pedestrian requests so no request is missed.
- To ensure safe switching between traffic lights
- To verify the design using a testbench and simulation.

3. System Description

3.1 Inputs

- **clk**: Clock signal that drives the FSM.
- **reset**: Resets the system to the initial state.
- **ped_req**: Pedestrian request button input.

3.2 Outputs

- **A_red, A_yellow, A_green**: Traffic lights for Road A.
- **B_red, B_yellow, B_green**: Traffic lights for Road B.
- **Ped_green**: Pedestrian crossing signal.

4. Finite State Machine (FSM) Design

4.1 FSM States

The FSM consists of five states:

State Name	Description
A_G	Road A Green, Road B Red
A_Y	Road A Yellow, Road B Red
B_G	Road B Green, Road A Red
B_Y	Road B Yellow, Road A Red
PED	Pedestrian Crossing (All roads Red)

4.2 State Transitions

- The system starts in A_G after reset.
- Each state remains active for a fixed time controlled by a timer.
- After B_Y, the FSM checks if a pedestrian request is latched.

- If yes, it moves to PED state.
- If no, it returns to A_G.

- After the PED state, the system always returns to A_G.

5. Pedestrian Request Handling

To ensure that a pedestrian request is not missed:

- A signal called ped_latched is used.
- When ped_req is pressed, ped_latched becomes 1.
- The request stays stored until the FSM enters the PED state.
- Once the pedestrian crossing is served, the latch is cleared.

This ensures reliable and safe pedestrian operation.

6. Timer Operation

- A 4-bit timer is used to control how long each state stays active.
- The timer increments on every clock cycle.
- When the timer reaches 9, it resets to 0 and the FSM moves to the next state.

This approach provides equal and predictable timing for all signals.

7. Verilog Module Description

7.1 traffic_light.v (Main Design)

The main module contains:

- FSM state register
- Timer logic
- Pedestrian request latch
- Output logic based on the current state

The design uses:

- Sequential logic (always @(posedge clk)) for state, timer, and latch
- Combinational logic (always @(*)) for outputs

This separation makes the design clean, readable, and synthesizable.

8. Testbench Description

8.1 traffic_light_tb.v

The testbench is used to simulate and verify the design.

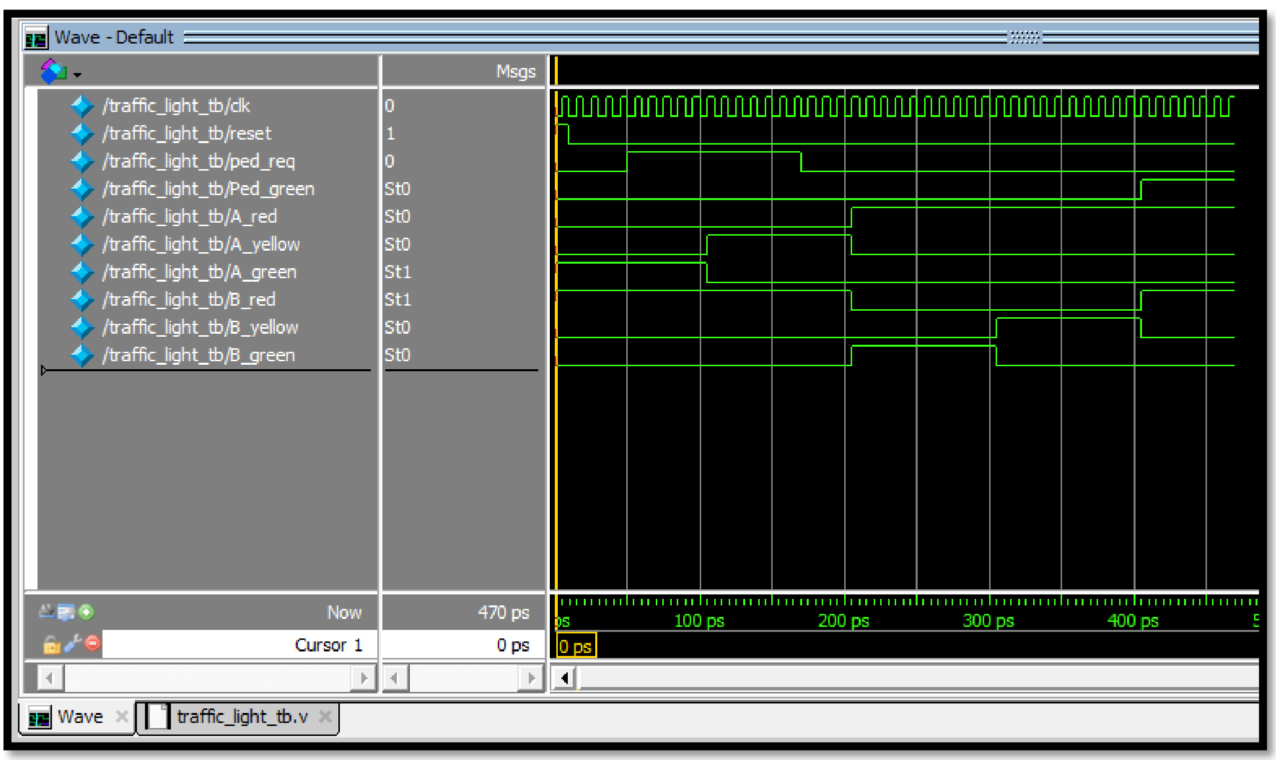
Functions of the testbench:

- Generates a clock signal (10 time units period).
- Applies reset at the beginning.
- Simulates a long pedestrian button press.
- Observes output signals during simulation.

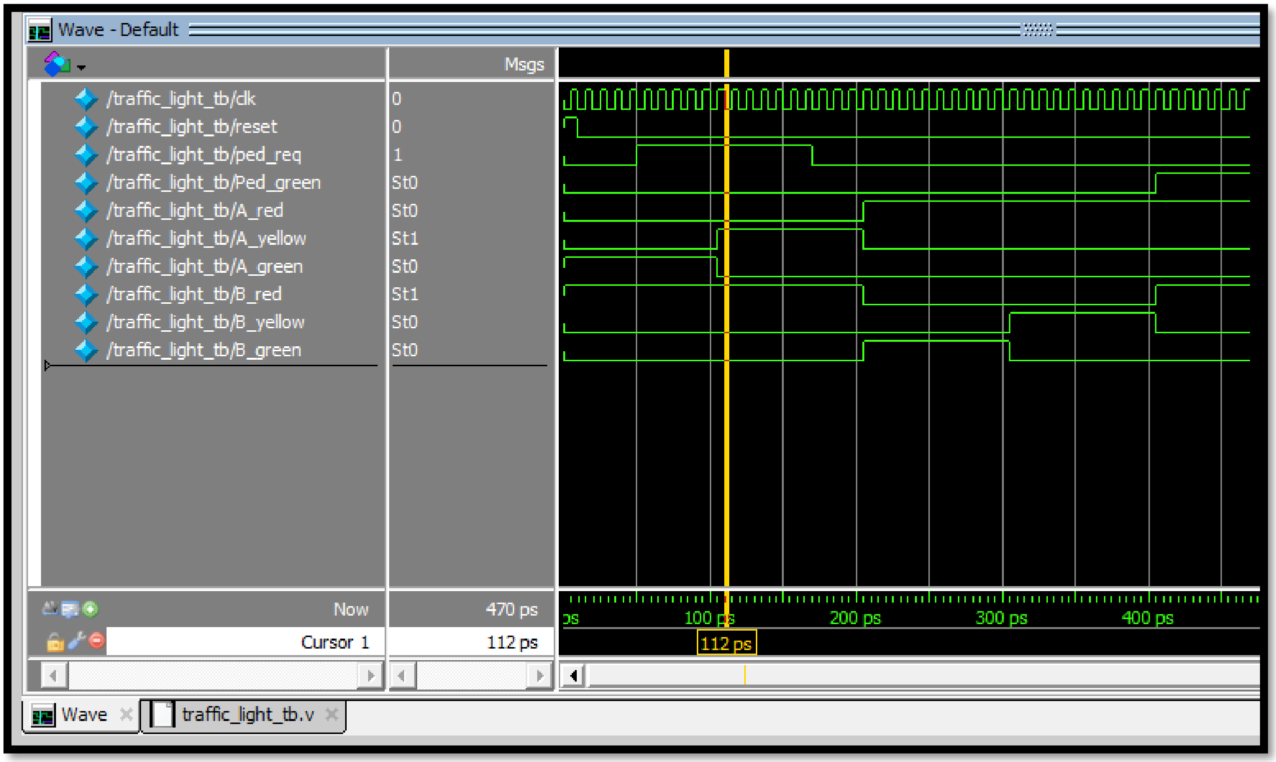
The pedestrian request is held high long enough to ensure it is latched and correctly served.

9. Simulation Results (Explanation)

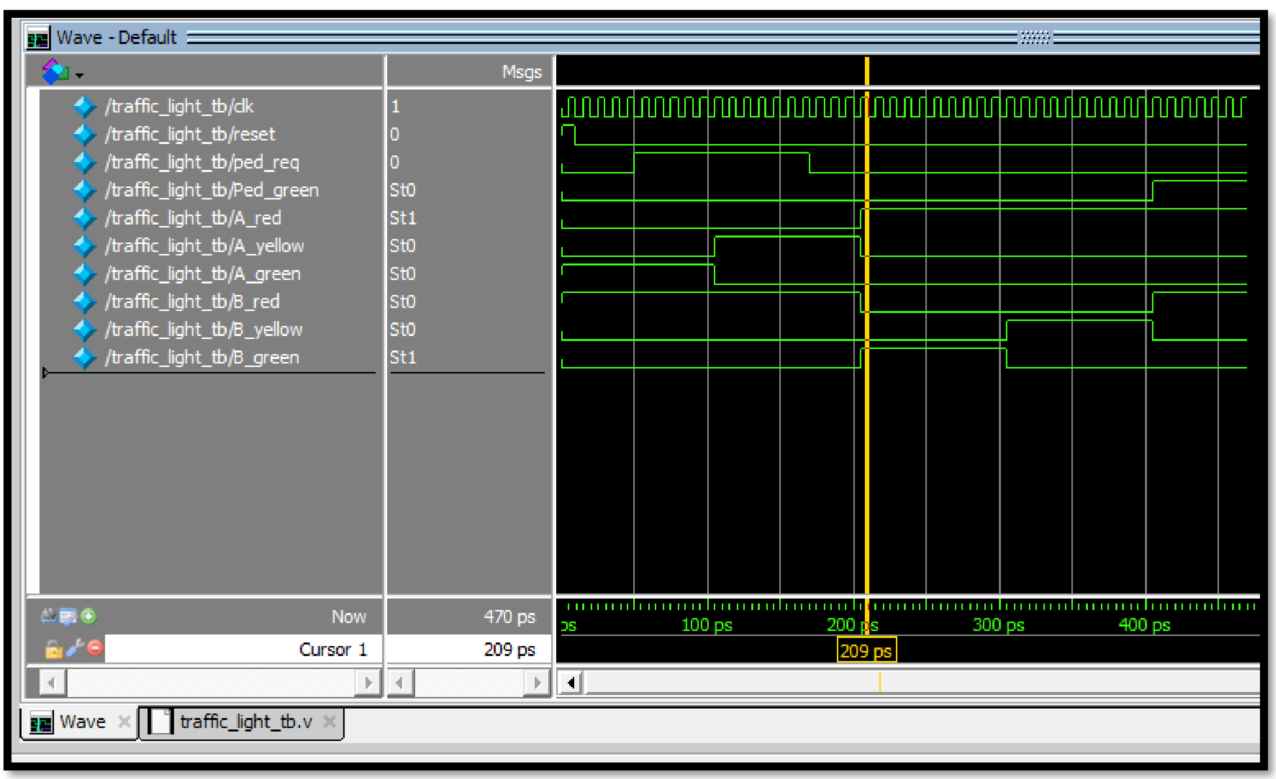
When road A is green, Road B is Red



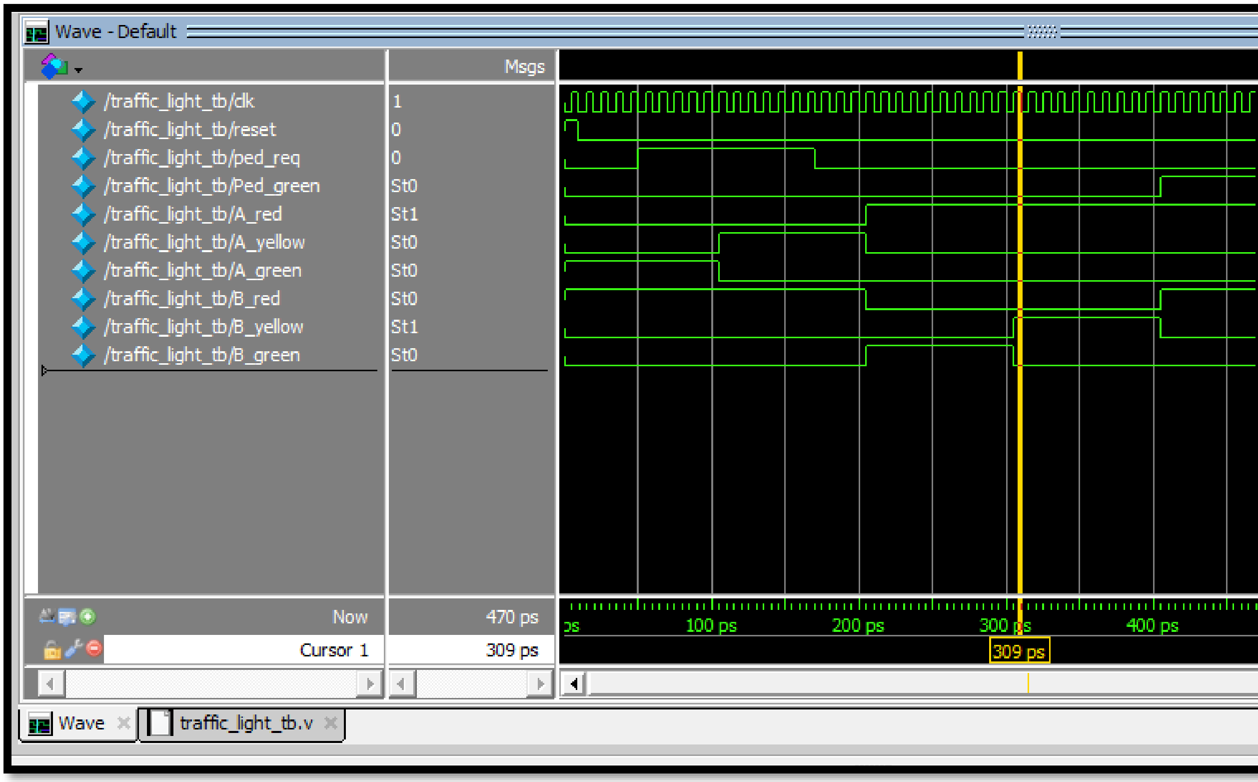
When Road A is yellow, Road B is Red



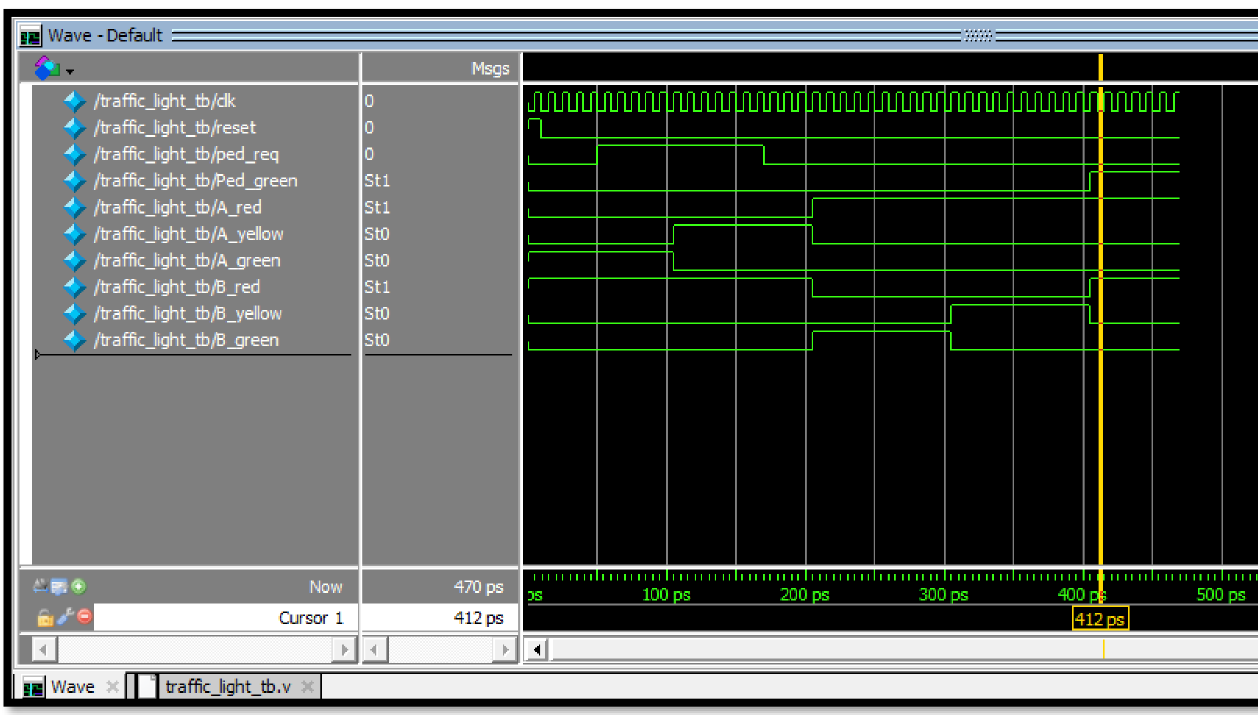
When Road A is Red, Road B is green



When A Road is Red, Road B is yellow



When Pedestrain is green, Road A and B are Red



From simulation, the following behavior is observed:

- After reset, Road A turns green.
- Lights switch correctly between green, yellow, and red.
- When the pedestrian button is pressed, the request is stored.
- After Road B yellow, the system enters pedestrian mode.
- During pedestrian mode, both roads show red and pedestrian signal turns green.
- After pedestrian crossing, the system resumes normal traffic flow.

This confirms correct FSM operation.

10. Advantages of the Design

- Simple and clear FSM-based approach.
- No pedestrian request is lost due to latching.
- Easy to expand (extra roads or sensors).
- Fully synthesizable and simulation-friendly.

11. Applications

- Road intersections
- School and hospital crossings
- Smart city traffic systems
- Embedded and FPGA-based controllers

12. Limitations

- Fixed timing for all states.
- No emergency vehicle handling.
- Single pedestrian crossing only.

13. Future Enhancements

- Adaptive timing based on traffic density.
- Emergency vehicle priority.
- Countdown timers for pedestrians.
- Multiple pedestrian request buttons.

14. Conclusion

This project successfully demonstrates the design and implementation of a Traffic Light Controller with Pedestrian Mode using Verilog. By using FSM concepts, timer-based control, and request latching, the system ensures safe and efficient traffic management. The simulation results verify correct operation, making this design suitable for academic and practical applications.